

May 11, 2026

Dr. Peter H. Thrall
Editor-in-Chief, *Ecology Letters*
CSIRO National Collections & Marine Infrastructure
GPO Box 1600, Canberra ACT 2601, Australia

Dear Dr. Thrall,

We are pleased to submit our manuscript, “*Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses*”, for consideration as a Letter in *Ecology Letters*.

Understanding when and why mutualisms break down under environmental stress is a central challenge in ecology, with increasing urgency as climate change pushes species into novel conditions. Despite extensive theory on the stress dependence of symbiotic benefits, rigorous empirical tests remain rare, particularly those that move beyond single host species or controlled settings to evaluate symbiont effects across the full range of conditions hosts naturally encounter. Our study addresses this gap through an integrative design. We conducted common garden experiments with three grass species hosting vertically transmitted fungal endophytes (*Epichlöe* spp.) across a natural aridity gradient spanning seven sites in the south-central United States, with herbivore exclusion treatments nested within sites. We analyzed demographic vital rates using hierarchical models that partition variation across species, sites, and treatments, providing resolution unattainable in simpler designs. A complementary greenhouse experiment, which independently manipulated soil origin, allowed us to isolate mechanisms underlying field patterns. By combining multi-species field experiments, demographic modelling, and controlled greenhouse tests, our study provides a rare, mechanistic evaluation of context dependent symbiosis. We believe this approach allows us to address both generality and underlying drivers, questions of broad relevance to the readership of *Ecology Letters*.

We have opted out of double-blind review because this paper builds upon our previous work, so our identities could be easily deduced. Our identities are also evident from our GitHub repository, where we direct reviewers to find all data and code necessary to reproduce our analyses.

Sincerely yours,

Jacob K. Moutouama
On behalf of all co-authors

Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses

Jacob K. Moutouama  *^{1, 2}, Josh C. Fowler³, Akiem Gough⁴, Julia Martin⁴, Chris Oxley⁴, Karl Schrader⁴, Karis Williams⁴, and Tom E.X. Miller ⁴

¹Department of Botany, University of British Columbia, Vancouver, BC, Canada

²African Centre for DNA Barcoding, University of Johannesburg, Johannesburg, South Africa

³Department of Environmental Studies, University of Colorado Boulder, Boulder, CO, USA

⁴Department of BioSciences, Rice University, Houston, TX, USA

Running Title: Stress and Grass-Endophyte Symbioses

Keywords: demography, parasitism, plant-fungi interactions, range limits, stress gradient hypothesis, symbiosis

Article Type: Letter

Data accessibility statement: Upon acceptance, all data and computer scripts supporting the results will be archived in a Zenodo package. During peer review, all data files required to reproduce the analyses, including annual census data, are temporarily hosted via Dropbox and available at <https://www.dropbox.com/scl/fo/fgxng2mcemjyv41bihul9/AGiZs5mPRGcCZURpGyvhjCE?rlkey=7mgdyrus3ga44yj3ygfncfmd&dl=1>. All analysis scripts (Stan and R) are available at <https://github.com/jmoutouama/endo-range-limits>.

Conflict of interest statement: None.

Acknowledgments This research was supported by National Science Foundation Division of Environmental Biology award 2208857. We gratefully acknowledge the individuals and institutions who hosted our geographically-distributed experiment, including the University of Louisiana, Sam Houston State University, Texas A&M University, and University of Texas at Austin. JKM was supported by start-up research funds GR036071 from the University of British Columbia and by NSERC grants GR038661 and GR038746.

Authorship statement: J.K.M. contributed to data collection, data analysis, and led manuscript drafting. J.C.F., A.G., J.M., C.O., K.S., and K.W. contributed to data collection and manuscript revisions. T.E.X.M. contributed to research conception, data collection, data analysis, and manuscript revisions.

This file contains: Abstract (150 words), Main Text (4745 words), Figures (1-6), Supporting Information - Supplemental Methods, Supplemental Figures S1-S10, References (72).

*Corresponding author: jmoutouama@gmail.com